

TABLEAU – LEVEL OF DETAIL

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Level of Detail

- Level of detail is a syntax that simplifies and extends Tableau's calculations by making it possible to address detailed queries directly.
- LOD represents an elegant and powerful way to answer questions involving multiple levels of granularity in a single visualization.
- Structure of LOD:

**{[scoping keyword] <dimension
declaration>: <aggregation expression> }**

Include LOD Calculations

- Include LOD calculations allow you to include specific dimensions while performing calculations.
- By using the 'INCLUDE' keyword, you can specify which dimensions you want to include in your calculation.
- This is helpful when you want to perform calculations only for specific dimensions, ignoring the others in your visualization.



Exclude LOD Calculations

- Exclude LOD calculations allow you to exclude specific dimensions while performing calculations.
- By using the 'EXCLUDE' keyword, you can specify which dimensions you want to exclude from your calculation.
- This is useful when you want to perform calculations for most dimensions but exclude certain dimensions from the calculation.



Fixed LOD Calculations

- allow you to specify a fixed level of detail in your calculation.
- By using the 'FIXED' keyword, you can ensure that the calculation is performed at a specific level of detail, regardless of the dimensions in your visualization.
- This is helpful when you want to perform calculations at a consistent level, regardless of any filters or dimensions applied.



Type of LOD	Description	Example
Include Calculation	Computes a calculation at a specified level of detail, aggregating all other dimensions.	{INCLUDE [Dimension]: SUM([Measure])}
Exclude Calculation	Excludes a dimension from a calculation's level of detail, aggregating other dimensions.	{EXCLUDE [Dimension]: AVG([Measure])}
Fixed Calculation	Computes a calculation at a fixed level of detail, independent of visualization context.	{FIXED [Dimension]: MAX([Measure])}

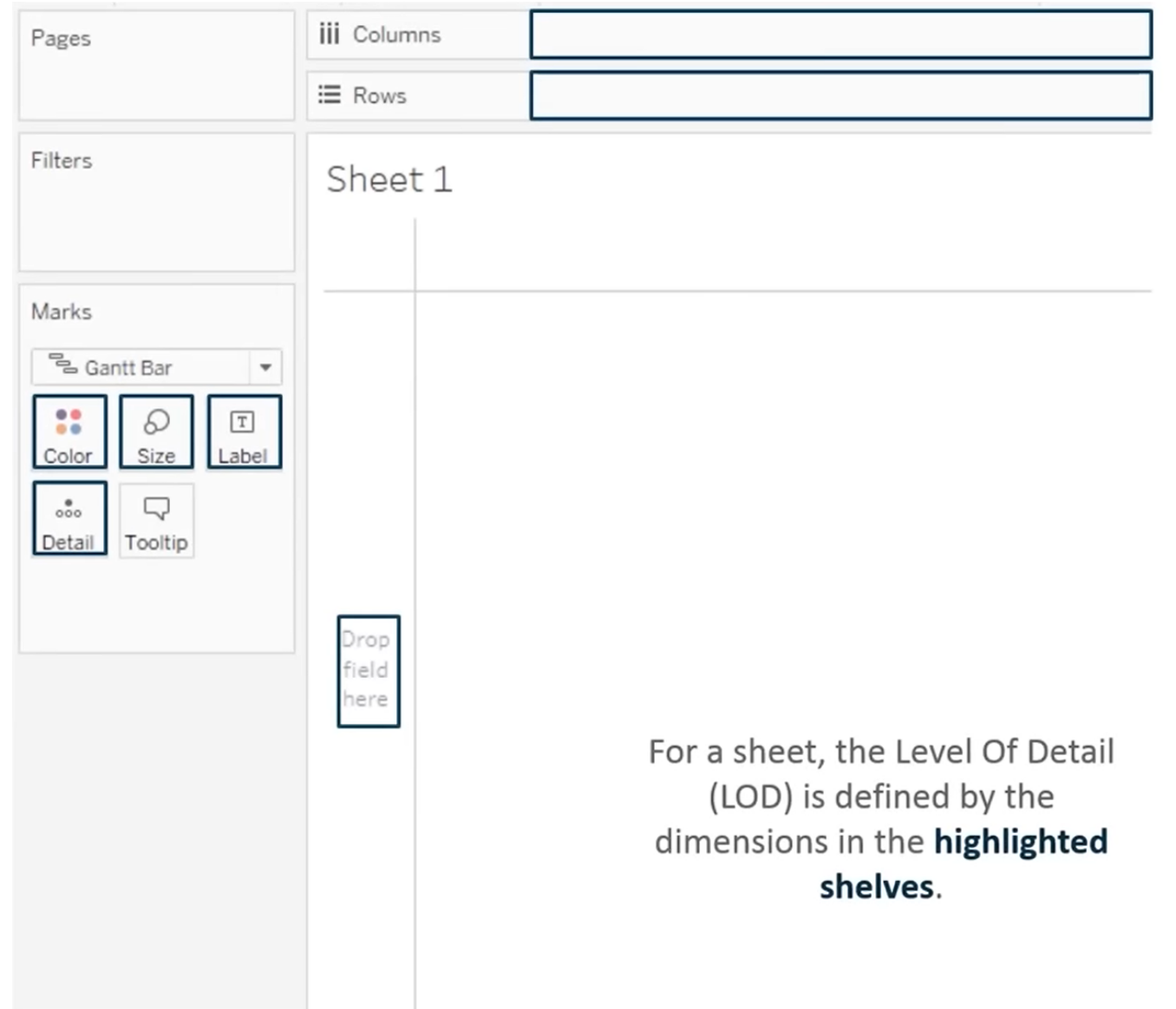
Level of Detail (LOD) Expressions

- Level of Detail (LOD) expressions in Tableau allow you to control the granularity of your analysis.
- LOD expressions can contain aggregate functions (SUM, AVG, MIN, MAX, etc.)
- They enable you to perform calculations at a specific level of detail, regardless of the dimensions in your view.
- LOD expressions can be used to calculate values at different levels, such as across the entire dataset, at the level of a specific dimension, or at a combination of dimensions.

Creating LOD Expressions

- To create an LOD expression, start with the LOD keyword followed by the desired level of detail.
- Next, use curly braces to enclose the fields and aggregations you want to include in the expression.
- Here's an example of a basic LOD expression:
- LOD expression: {FIXED [Category] : SUM([Sales])}
- This expression calculates the sum of sales for each category, fixing the level of detail to the category field.

Areas Affecting LOD Expressions



The screenshot displays the Tableau interface for configuring a Level of Detail (LOD) expression. On the left, the 'Marks' shelf is highlighted with a blue border. Below it, the 'Detail' shelf is also highlighted. The 'Columns' and 'Rows' shelves are empty. The main view area shows 'Sheet 1' with a grid. A small box labeled 'Drop field here' is visible on the left side of the grid.

Pages

Columns

Rows

Filters

Marks

Gantt Bar

Color

Size

Label

Detail

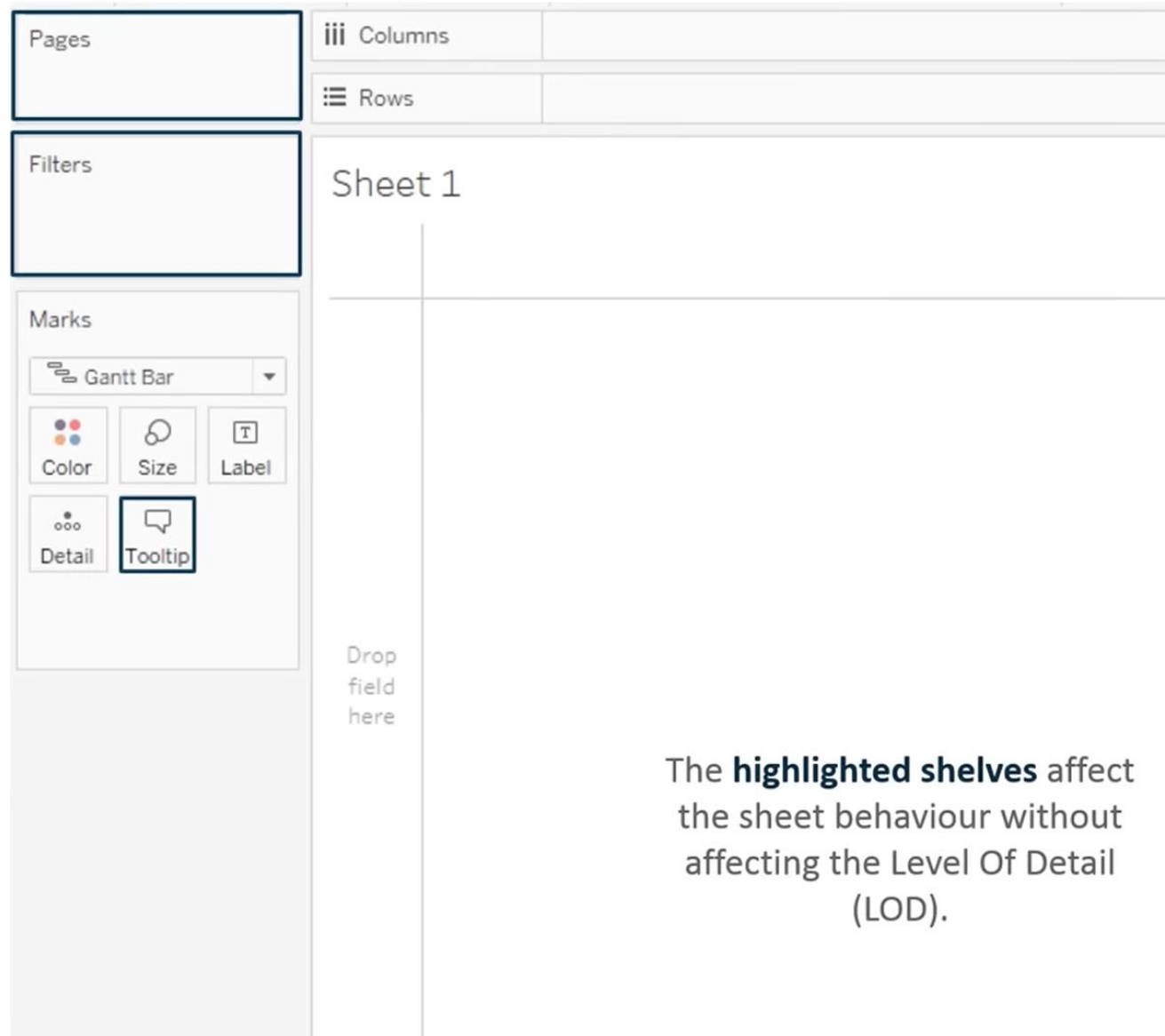
Tooltip

Drop field here

Sheet 1

For a sheet, the Level Of Detail (LOD) is defined by the dimensions in the **highlighted shelves**.

Areas Which Do Not Affect LOD Expressions



The screenshot shows the Tableau interface with the following elements highlighted:

- Pages** shelf in the top left.
- Filters** shelf in the top left.
- Marks** card with the following options:
 - Gantt Bar** (selected in the dropdown).
 - Color** (represented by a color icon).
 - Size** (represented by a size icon).
 - Label** (represented by a text icon).
 - Detail** (represented by a detail icon).
 - Tooltip** (represented by a tooltip icon and highlighted with a blue border).

The main view area is labeled "Sheet 1" and contains a large empty space with the text "Drop field here" at the bottom left.

The **highlighted shelves** affect the sheet behaviour without affecting the Level Of Detail (LOD).

Advantages of Using LOD Expressions



LOD expressions provide flexibility and control over calculations in Tableau.



They allow you to create custom aggregations and dimensions without relying solely on the built-in functions.



Using LOD expressions, you can easily handle complex data analysis scenarios that are otherwise difficult to achieve.



They help you gain deeper insights and answer more complex business questions.

Thank You



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